

Analysis Report — Camera-Based LKA System

1. System Overview

The implemented system is a:

monocular camera-based Lane Keeping Assist (LKA)

developed using:

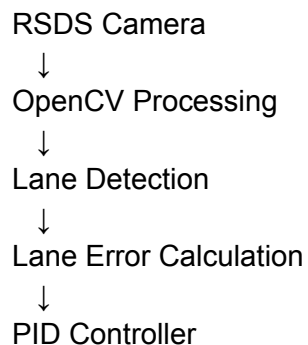
- IPG CarMaker,
- RSDS camera streaming,
- Python,
- OpenCV,
- PID steering control,
- DVA steering actuation.

The system performs:

- real-time lane detection,
- lane error estimation,
- steering correction,
- closed-loop lane tracking.

2. Control Architecture

The LKA pipeline follows:



↓
DVA Steering Control
↓
Vehicle Motion

The controller uses:

camera-only perception for steering control.

Road sensor data (`Vhc1.tRoad`) is used only for:

- evaluation,
 - metrics,
 - performance analysis.
-

3. Perception Pipeline Analysis

Image Processing

The OpenCV pipeline includes:

- grayscale conversion,
- Gaussian blur,
- Canny edge detection,
- Region of Interest (ROI) extraction.

This reduces:

- image noise,
 - unnecessary computation,
 - non-road features.
-

Lane Detection

The system uses:

Probabilistic Hough Line Transform

to detect lane-line segments.

Only:

left-lane tracking

is used.

This approach was selected because:

- midpoint lane estimation produced instability,
- perspective distortion affected center estimation,
- single-lane tracking gave more stable curved-road behavior.

4. Lane Error Estimation

Lane error is calculated using:

```
error_px = (  
    DESIRED_LEFT_X - left_lane_x  
)
```

where:

- `DESIRED_LEFT_X = 120`
acts as the calibrated visual lane reference.

The error is normalized and clipped before being sent to the controller.

This represents:

visual lane deviation in image space.

5. PID Controller Analysis

The steering controller uses:

PID-based lateral control.

Controller gains:

Parameter	Value
KP	0.80
KI	0.0005
KD	0.10

The controller continuously computes steering corrections based on:

- proportional lane deviation,
 - accumulated drift,
 - rate of lane-error change.
-

6. Steering Behavior Analysis

The controller generates:

- positive steering for one-direction correction,
- negative steering for opposite-direction correction.

Steering smoothing:

STEER_SMOOTHING = 0.75

reduces:

- abrupt steering changes,
- oscillation,
- unstable lateral motion.

Small steering oscillations were observed even on straight roads due to:

- camera noise,
- Hough line variability,
- continuous closed-loop correction.

This behavior is expected in real-time LKA systems.

7. Curved-Road Performance Analysis

The system was tested on:

- straight roads,
- curved-road sections,
- continuous curvature scenarios.

Observed Behaviour

- Lane error increased during curve entry.
- Steering magnitude increased accordingly.
- The PID controller continuously corrected the vehicle trajectory.
- The ATV successfully maintained lane-following behavior.

The system demonstrated:

stable closed-loop lane tracking during continuous curvature.

8. Greatest Deviation Zones

The largest lane deviations occurred during:

- transition from straight road to curve,
- sharp curvature entry,
- sustained turning regions.

These deviations occurred because:

- lane geometry changed rapidly,
 - perspective distortion increased,
 - lane position shifted continuously in image space.
-

9. Metrics Analysis

The system computes:

- RMSE,
- ILC (In-Lane Compliance),
- Maximum Deviation,
- DNF status.

However:

Vhc1.tRoad does not represent true camera-based lane-center error.

It represents:

- OpenDRIVE road-reference geometry.

Therefore:

- steering control remains camera-only,
 - road sensor data is used only for evaluation.
-

10. Key Strengths of the System

The implementation successfully demonstrates:

- ✓ Real-time camera perception
 - ✓ OpenCV lane detection
 - ✓ PID-based steering control
 - ✓ Closed-loop lateral control
 - ✓ Curved-road lane keeping
 - ✓ Camera-only steering actuation
 - ✓ Stable runtime behavior
-

11. Limitations Identified

The following limitations were observed during testing:

Limitation	Cause
Steering bias	fixed visual reference calibration
Perspective distortion	monocular camera geometry
Increased curve deviation	rapid lane-position changes
Small steering oscillations	continuous feedback correction
Sensitivity to lane markings	classical CV approach

12. Lessons Learned

Testing showed that:

- single-lane tracking improves curved-road stability,
- PID tuning significantly affects steering smoothness,
- steering smoothing is important for stable lateral control,
- camera-only lane tracking is more challenging on continuous curves.

The project also demonstrated that:

a monocular camera can successfully perform real-time lane-following control in simulation.

13. Proposed Future Improvements

Possible future improvements include:

- bird's-eye-view transformation,
- adaptive lane reference estimation,
- polynomial lane fitting,
- dynamic PID tuning,
- Kalman filtering,
- sensor fusion,
- deep-learning lane segmentation.

14. Final Conclusion

The developed system successfully demonstrates:

a functional camera-based Lane Keeping Assist system

using:

- RSDS camera streaming,
- OpenCV lane detection,
- PID steering correction,
- DVA-based vehicle actuation.

The controller performs:

- real-time lane tracking,
- autonomous steering correction,
- stable lane-following behavior

without relying on:

- road sensors,
- lane sensors,
- OpenDRIVE geometry for steering control.